

4.0 DOWN SYNDROME



Down syndrome is a genetic condition where people are born with an extra chromosome. Most people have 23 pairs of chromosomes within each cell in their body, for a total of 46. A person diagnosed with Down syndrome has an extra copy of chromosome 21, which means their cells contain 47 total chromosomes instead of 46. This changes the way their brain and body develop. Down syndrome, also called *trisomy 21*, happens by chance, can't be prevented, and isn't caused by anything a parent did or didn't do. Fortunately, the health problems can be managed well, especially when found early, and many resources are available to help kids and their families.

4.1 What causes Down syndrome in a child?

When a baby is conceived, a normal egg cell and normal sperm cell start with 46 chromosomes. The egg and sperm cells then divide in half. The egg and sperm cells then have 23 chromosomes each. When a sperm with 23 chromosomes fertilizes an egg with 23 chromosomes, the baby will then have a complete set of 46 chromosomes. Half are from the father and half are from the mother.

But sometimes an error occurs when the 46 chromosomes are being divided in half. An egg or sperm cell may keep both copies of chromosome number 21, instead of just 1 copy. If this egg or sperm is fertilized, then the baby will have 3 copies of chromosome number 21. This is called trisomy 21. Sometimes the extra number 21 chromosome or part of it is attached to another chromosome in the egg or sperm. This may cause ***translocation Down syndrome***. This is the only form of Down syndrome that may be inherited from a parent.

A rare form is called mosaic trisomy 21. This is when an error in cell division happens after the egg is fertilized. People with this syndrome have both normal cells and some cells with an extra chromosome number 21.

4.1.1 Physical signs of Down syndrome

Physical signs of Down syndrome are usually present at birth and become more apparent as your baby grows. They can include:

- A flat nose bridge.
- Slanted eyes that point upward.
- A short neck.
- Small ears, hands and feet.
- Weak muscle tone at birth.
- Small pinky fingers that points inward towards the thumb.
- One crease in the palm of their hand (palmar crease).
- Shorter-than-average height.

As your child grows, additional symptoms can arise because of the way that their body developed in the uterus, including:

- Ear infections or hearing loss.
- Vision problems or eye diseases.
- Dental problems.
- Being more prone to infections or illnesses.
- Obstructive sleep apnea.

- Congenital heart disease.

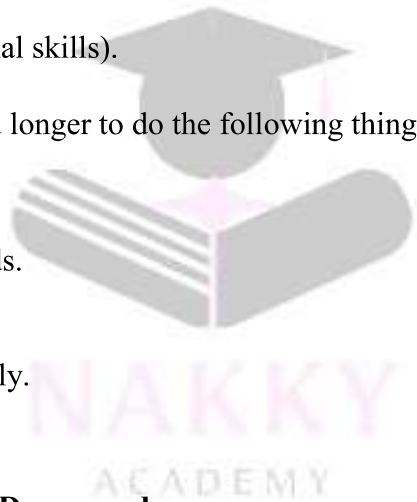
4.1.2 Cognitive symptoms of Down syndrome

The child with Down syndrome may have cognitive development challenges as a result of their extra chromosome. This can cause intellectual or developmental disabilities. The child's ability to meet developmental milestones, or things the child can do at a certain age, may differ from other children, including how they:

- Walk and move (gross and fine motor skills).
- Speak (language development skills).
- Learn (cognitive skills).
- Play (social and emotional skills).

As a result, it may take the child longer to do the following things:

- Toilet training.
- Speaking their first words.
- Taking their first steps.
- Eating food independently.



4.1.3 Behavioral symptoms of Down syndrome

A child diagnosed with Down syndrome may exhibit behavioral symptoms. This can be the result of the child not being able to communicate their needs to you or their caregivers effectively. Behavioral symptoms of Down syndrome could include:

- Stubbornness and tantrums.
- Difficulty paying attention.
- Obsessive or compulsive behaviors.

4.2 Taking Care of Children with Down Syndrome.

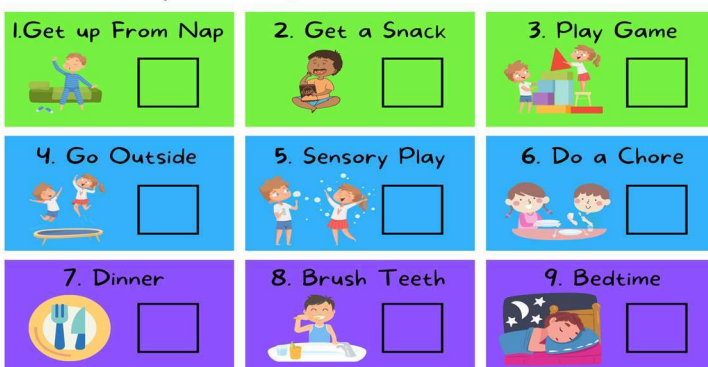
- **Give simple, clear directions:** Language is often difficult for children with Down syndrome. The more complicated your speech, the less likely the child is to do what you want. So directions should be specific, directive (a request, not a question), and contain the fewest steps possible.

For example, say: “Brush your teeth now, please.” Don’t say: “Can you please go upstairs and brush your teeth before we have to leave for school?”

Say: “Please put your pajamas on.” Don’t say: “I already told you it’s time to get ready for bed! If you put on your pajamas, I’ll come read you a story.” You may wish to speak with your child’s school speech pathologist or with a private provider about other options, such as sign language or the use of a picture exchange system.

- **Establish a routine, and stick to it:** Every morning, most adults do their routine in the same order. For example: use the bathroom, take a shower, get dressed, have breakfast, brush teeth, get lunch ready to take to work. Having a routine makes life easier! The same is true for children with Down syndrome, but routine is even more important. The child is likely to do best when the day’s structure is the same as it was the day before. Try your best to make a routine and to help your child understand what that routine is.
- **Use visual schedules:** Many schools use this approach. Here is an example of an afterschool routine shown visually. It tells a child that after she gets home from school, she needs to hang up her coat, have a snack, play, read a book, and then eat dinner. This chart was easy to make and is easy to follow. Something like this might help your child move more smoothly through his day without so much work on your part.

My Night Routine



- **Plan for difficult Situations:** We all do better with structure, but we can't always keep things the same. As a parent or caregiver, you know your child best. You probably are aware of some things that are difficult for your child. Anticipating these events can be helpful and you can help your child prepare for changes in order to reduce his or her worry or behavior problems. One approach to getting ready for what's coming is called Social Stories. This technique involves outlining coming events for your child using a book format. This is often through pictures, but may also involve some words. Social stories can help your child prepare for difficult or complex events such as having blood drawn, going to a new classroom, or having a birthday party.
- **Make time for the fun stuff:** When children are having behavior problems, they tend to get lots of negative attention. It is important to have positive interactions with your child, even when things are at their worst. Make time each day to play a game, read a book, draw, cook a fun dish, or watch a video with your child, even if things have been hard around the house.
- **Reward good behavior:** Set up a system to reward the behaviors you want to encourage. Start by answering these questions:
 - What are two or three things you would like your child to do more often?
 - Could your child do these things if he or she were willing?
 - Does your child need to be rewarded right away to understand the connection between completing a job and getting a prize?
 - Can he understand being rewarded later on for something he did earlier in the day or week? (This is quite difficult to do!)
 - What does your child like that could be used to reward? Stickers? Poker chips? Quarters to buy something later on? Time to play a game with someone special?Choose a few behaviors that are important to you, and use a chart to help your child achieve them (see below).
- **Making a good-behavior chart:** Children with Down syndrome often respond to visuals better than to being told what to do. Make a chart with pictures of what you'd like your child to do and put it up where he or she can see it, maybe on the bedroom wall or refrigerator. You can draw the pictures, cut them out of magazines, take photos, or

photocopy them from books. When you're starting out, make the goals really simple and positive. We want your child to like this so he keeps trying! It's okay to help out at first, but encourage your child to do jobs independently to earn a prize after he has tried the chart out for a few days. As you go, add in some more difficult jobs to go with the ones your child has mastered. Remember, go slowly and keep it simple.

Be consistent with your use of the chart. This will not work if you stop doing it when it's going well (or not so well). If you are having trouble keeping it going or are not having success, troubleshoot with a professional to figure out why.



		Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Great getting ready in the morning!								
Good at School! Good at home!								
Eating Dinner Nicely!								
Great Bedtime!								
WOW! Bonus								
TOTAL								

- Pick your battles:** Wouldn't it be great if we could make everyone in our lives behave exactly as we'd like them to? With other adults, you know that you can't always get what you want. The same is true of your child. A lot of arguments, standoffs, and heartache can be avoided by simply choosing your battles. It is often beneficial to step back and consider whether a certain behavior is worth reacting to. Consider the following when determining which battles to pick:

- Is the behavior dangerous? If so, then you need to intervene.
 - If the behavior is not dangerous, is it one of the two or three behavior problems that you'd most like to decrease? If not, you might want to let it go...for now.
 - If the behavior is not dangerous but is just bothersome, you might be best to ignore it. Not only will you save yourself some headaches, but this might even get rid of the behavior.
- **Avoid power struggles BY OFFERING CHOICES AND EMPOWERING YOUR CHILD!** Power struggles are very common between parents and children. A power struggle means that a child wants to do one thing, and a parent wants her to do another. And nobody wants to give in. Power struggles can be very upsetting, and are often very hard on both parents and children. They can be particularly hard for children with Down syndrome, who are often social and affectionate and may find it very upsetting to be fighting with a loved one. When thinking about power struggles, keep these things in mind:
 - Just like you, your child would like some control over his or her life.
 - You are the boss, but people tend to like bosses who listen to them and given them some power.
 - When you offer people even a little bit of control, they tend to feel better. To avoid a power struggle, you might try the following:
 - Offer your child a choice. If there are three things that need to be done, allow your child to decide in what order he or she will complete those things.
 - Provide your child with an “if-then” option. If she does what you want first, she can do something of her choosing next.
- **Disarm and Distract:** Picture a two-year-old boy who is crying because his pacifier was taken away. This child is too young to respond to most forms of punishment, and this behavior is not something that should be punished anyhow. Many parents, whether they know it or not, use the “disarm and distract” principle. That is, they would simply give this toddler a new toy to help him stop crying. This same principle can be used to manage behavior problems or strong emotional reactions in older children and children with Down syndrome. Forget about the old behavior, or whatever triggered it, and introduce something new to help your child calm down and shift her focus.

- **THE POWER OF IGNORING:** Many children with Down syndrome are very social. They often love to receive attention, even if it is negative. If a behavior is not unsafe, try ignoring it and giving no feedback. Sometimes this is enough to get rid of a behavior!
- **Take away privileges:** Another way to discipline effectively is to take things away from your child when he or she misbehaves. For example, if your son hits his sister, he may lose TV time. Keep in mind that discipline is only going to work if your child understands it. Therefore, you should only take something away from your child for misbehavior if he or she understands the reason for this.

